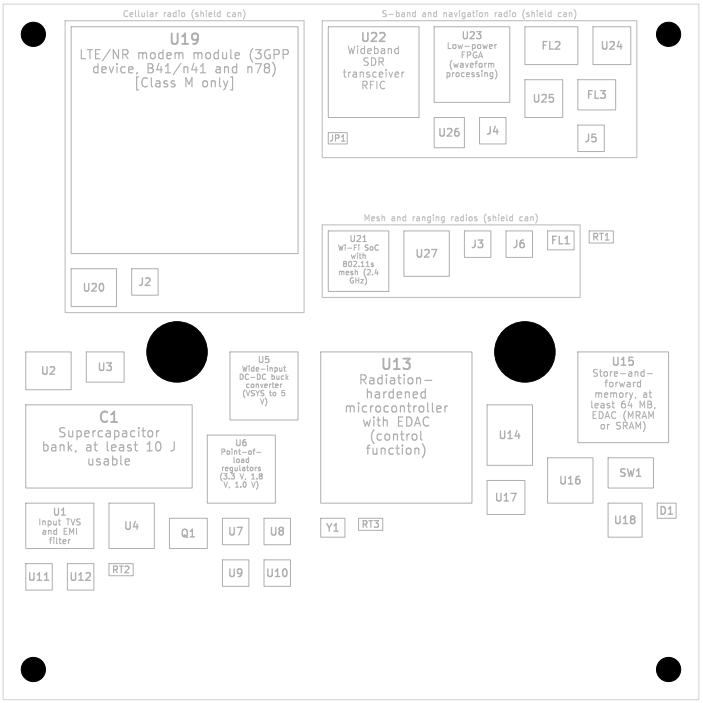


PART TYPES (generic placeholders, no part numbers)

J1 Host connector, module half: contact pads and guide bushings (PBS-HW-CON-01) (back)
MH1-4 Mounting hole, chassis bonded
U1 Input TVS and EMI filter
U2 Hot-swap controller with inrush limit
U3 Swap-detect conditioning (comparator)
U4 Hold-up controller (ideal diode and supercapacitor charger)
C1 Supercapacitor bank, at least 10 J usable
U5 Wide-input DC-DC buck converter (VSYS to 5 V)
U6 Point-of-load regulators (3.3 V, 1.8 V, 1.0 V)
U7 Voltage supervisor with watchdog
U8 Current-limited load switch (eFuse), latch-up protection: cellular
U9 Current-limited load switch (eFuse), latch-up protection: mesh
U10 Current-limited load switch (eFuse), latch-up protection: S-band and navigation
U11 Current-limited load switch (eFuse), latch-up protection: ranging
U12 Current-limited load switch (eFuse), latch-up protection: S-band amplifier
U13 Radiation-hardened microcontroller with EDAC (control function)
U14 Radiation-tolerant NOR flash (two firmware images)
U15 Store-and-forward memory, at least 64 MB, EDAC (MRAM or SRAM)
U16 Single-pair Ethernet PHY (100BASE-T1)
U17 RS-422 transceiver
U18 Hot-swap I2C bus buffer with tag supply switch
Y1 TCXO reference oscillator
D1 Ready indicator LED
SW1 Release-request switch (latch handle)
U19 LTE/NR modem module (3GPP device, B41/n41 and n78) [Class M only]
U20 eSIM (MFF2) [Class M only]
J2 Coaxial antenna port: cellular
U21 Wi-Fi SoC with 802.11s mesh (2.4 GHz)
FL1 2.4 GHz band-pass filter (2400-2483.5 MHz)
J3 Coaxial antenna port: 2.4 GHz mesh
U22 Wideband SDR transceiver RFIC
U23 Low-power FPGA (waveform processing)
U24 S-band power amplifier, 2 W [SCM-L only]
JP1 RF bypass link (SCM-S: fitted in place of the amplifier) [SCM-S only]
U25 S-band front end: T/R switch and LNA
FL2 S-band duplexer (2025-2110 / 2200-2290 MHz)
U26 RF antenna selector switch (lid antenna or host RF port)
J4 Coaxial antenna port: S-band lid patch
FL3 Navigation band SAW filter and LNA (2483.5-2500 MHz)
J5 Coaxial antenna port: navigation patch
U27 UWB ranging transceiver (optional, off by default) [Optional]
J6 Coaxial antenna port: UWB
Q1 Heater driver, low-side MOSFET
H1 Polyimide film heater, zone A (radios) (back)
H2 Polyimide film heater, zone B (power and control) (back)
RT1 Platinum RTD temperature sensor (radio zone)
RT2 Platinum RTD temperature sensor (power zone)
RT3 Platinum RTD temperature sensor (control zone)

Generated by PBS-HW-LIB/kicad/tools/generate.py
Placeholders are not routed; nets show the required interconnect
PBS-HW-SCM-01 open reference design; generic placeholders, no part numbers
Pale Blue Systems Foundation

Sheet:
File: pbs-scm-core.kicad_pcb

Title: PBS-HW-SCM-01 Surface Communication Module, core board

Size: A4	Date: 2026-10-08	**Rev: 0.3**
KiCad E.D.A. 9.0.9		Id: 1/1